

Q1:

Q18: Why TRST is optional in JTAG?

Ans:- TRST is an optional signal in JTAG as 'Test Logic Reset' state can also be entered by keeping TMS I/p active for atleast 5 clock cycles.

Q2:

Q19: JTAG FSM, whether it is mealy or moore? Justify.

[Answer is given based on my understanding Direct ans along with justification was not given anywhere as far as I have searched].

Ans:- As per IEEE 1149.1 std, ^{and TCLK} I/p: TMS is the I/P of FSM

and O/Ps: of FSM

capture IR	capture DR
shift IR	shift DR
update IR	update DR.

If FSM is in shift IR state, shift IR = 1
 " " " " capture IR state, capture IR = 1
 " " " " update IR state, update IR = 1

 " " " " capture DR state, capture DR = 1
 " " " " shift DR state, shift DR = 1
 " " " " update DR state, update DR = 1

Based on the value of (i/p) TMS, the transition of FSM happens.

All the above O/Ps depends only on the current state.

So JTAG FSM is a moore m/c.

(my ans and my justification)

Q3:

What is the default value/state of the JTAG pins: TDI, TDO, TMS?

Ans:- Default value of TMS and TDI is 1.
(pull up i.e. unless an external i/p driver, its value will be 1).

Reason:- If suppose TMS is connected to 0, it might go into some state.

So for safety reason, TMS is tied to 1 so that the TAP will be in the next state.

Bypass in is 1111

TDI is tied to 1 for safe state.

If suppose by chance if TMS value change, then it might get hanged in some unknown state.

But if TDI is tied to 1, then tap will be bypassed.

TDO of one chip goes as the TDI of another chip.

So default value of TDO is also 1.

(Reason same as for TDI).

Q4: Why is MBIST called Memory Built in Self-Test?

Why is Self-Test?

Ans: Given an algorithm, the BIST circuitry automatically applies the values onto memory according to the algorithm and the output of memory is also automatically compared with expected value.